Numerical Simulation and Analytical Validation of a Two-Impulse Hohmann Transfer from 7000 km to 12000 km Earth Orbit Using GMAT

Research-Style Technical Paper

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Prepared from the uploaded GMAT script, GMAT output log, and parameter-selection report

Abstract

This paper presents a GMAT-based numerical simulation and analytical validation of a classical two-impulse Hohmann transfer from a low Earth orbit with semi-major axis 7000 km to a higher near-circular Earth orbit of approximately 12000 km. The project uses a nearly circular initial orbit, a 28.5 degree inclination, simplified angular geometry, and two impulsive velocity increments applied in the local VNB frame. Analytical Hohmann equations predict a transfer semi-major axis of 9500 km, transfer eccentricity of 0.2631579, first burn of 0.93497844 km/s, second burn of 0.81612489 km/s, and total delta-v of 1.75110333 km/s. The GMAT run reproduces the expected behavior: after the first burn, the spacecraft enters an elliptical transfer orbit with semi-major axis 9500.3195 km and eccentricity 0.2632564; after propagation to apoapsis and the second burn, the spacecraft transitions into a near-circular target orbit with immediate semi-major axis 11991.6936 km and eccentricity 0.0013181. After the final coast, the reported orbit remains near the target altitude with semi-major axis 11989.9951 km and eccentricity 0.0008341. The study confirms that the model is physically consistent with Hohmann-transfer theory while also showing small deviations caused by nonzero initial eccentricity, finite numerical propagation, and GMAT osculating elements under a higher-order Earth gravity model.

Keywords: GMAT, Hohmann transfer, orbital mechanics, impulsive maneuver, Earth orbit, astrodynamics simulation, delta-v validation

1. Introduction

Orbit-raising maneuvers are among the most fundamental tasks in astrodynamics. A spacecraft often begins in a lower parking orbit and must be moved into a higher operational orbit using a limited propellant budget. For the ideal case of two coplanar circular orbits, the Hohmann transfer provides the reference maneuver because it connects the initial and target radii using a tangential transfer ellipse and two impulsive burns.

The objective of this project is to model, simulate, and validate a Hohmann transfer in NASA GMAT, the General Mission Analysis Tool. GMAT is suitable for this problem because it provides spacecraft definition, force-model control, impulsive burns, event-based propagation, graphical orbit visualization, and report-file generation in one environment. The specific mission is to transfer a spacecraft from a 7000 km Earth-centered orbit to a 12000 km target orbit using two velocity-direction burns.

Although the Hohmann transfer is analytically simple, implementing it in a numerical mission-analysis environment is valuable because it exposes the difference between ideal equations and propagated osculating orbital elements. The analytical solution assumes two-body dynamics, perfectly circular coplanar orbits, and instantaneous burns. The GMAT simulation uses a nearly circular initial orbit, a realistic inclination, a numerical propagator, and an Earth gravity model with zonal and tesseral structure. The result is therefore an excellent small-scale research case for connecting theory, numerical modeling, and simulation outputs.

2. Background and Theoretical Framework

A Hohmann transfer between two circular coplanar orbits uses an elliptical transfer orbit tangent to the initial orbit at periapsis and tangent to the final orbit at apoapsis. The first burn increases orbital energy and raises the apoapsis. The second burn is applied at apoapsis and circularizes the spacecraft into the higher orbit. For radius ratio 12000/7000 = 1.7143, the Hohmann transfer is inside the range where the two-impulse solution remains the standard minimum-fuel reference for circular-to-circular transfer.

For an initial radius r1, target radius r2, and Earth gravitational parameter mu, the transfer-orbit semi-major axis, eccentricity, burn magnitudes, and half-period are given by:

a_t = (r1 + r2) / 2

e_t = (r2 - r1) / (r2 + r1)

Delta-v1 = sqrt(mu/r1) [sqrt(2 r2/(r1 + r2)) - 1]

Delta-v2 = sqrt(mu/r2) [1 - sqrt(2 r1/(r1 + r2))]

t_transfer = pi sqrt(a_t^3 / mu)

Using mu = 398600.4418 km^3/s^2, r1 = 7000 km, and r2 = 12000 km, the expected transfer ellipse has a_t = 9500 km and e_t = 0.2631579. The first burn is 0.93497844 km/s, the second burn is 0.81612489 km/s, and the total delta-v is 1.75110333 km/s. The ideal coast from periapsis to apoapsis is 4607.51 s, or 76.79 min.

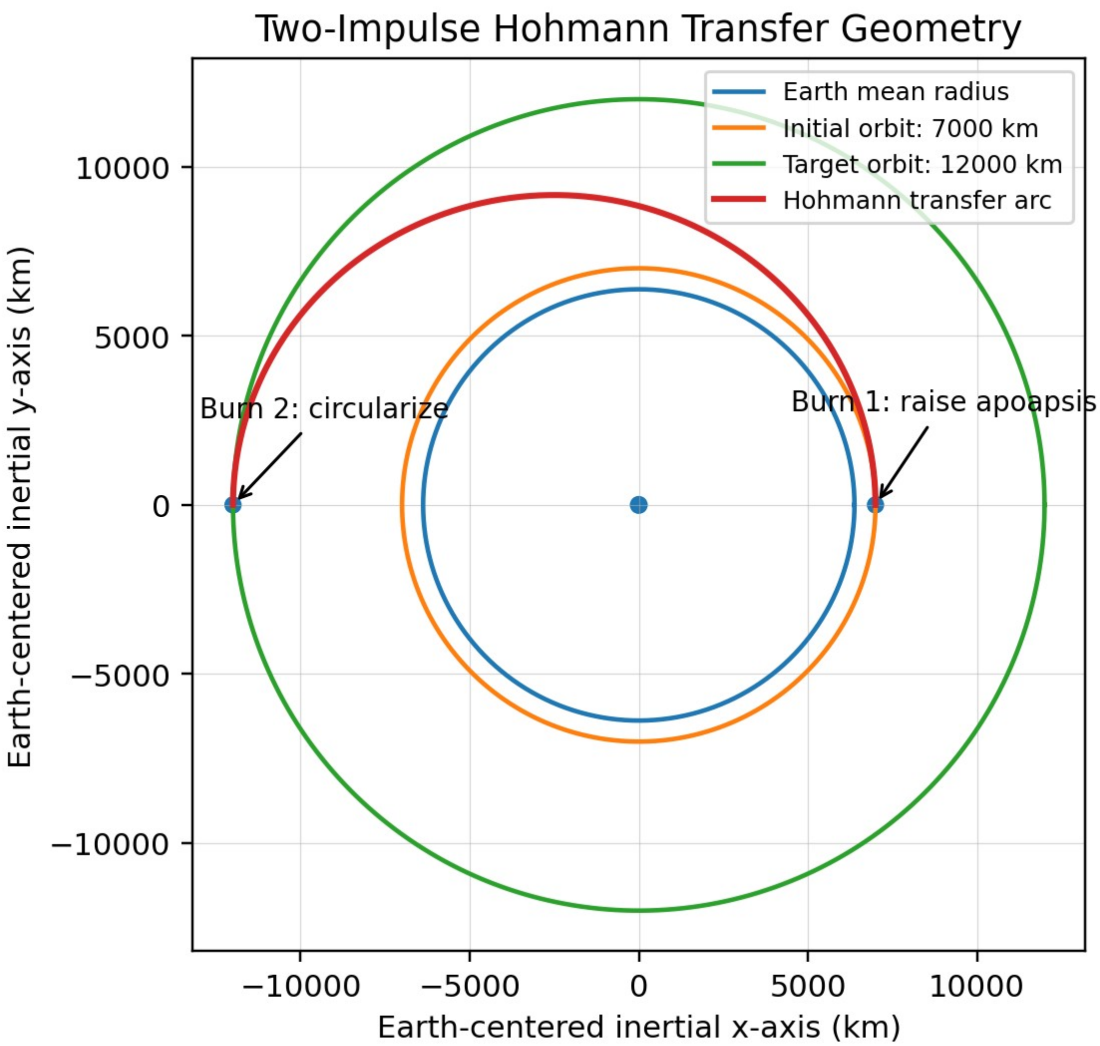

Figure 1. Analytical Hohmann-transfer geometry used to interpret the GMAT mission sequence.

3. GMAT Simulation Architecture

The simulation was constructed as a scripted GMAT mission sequence. The spacecraft, named Sat, was initialized in EarthMJ2000Eq coordinates with Keplerian display elements. The model begins at semi-major axis 7000 km, eccentricity 0.0001, inclination 28.5 degrees, and zero values for RAAN, argument of periapsis, and true anomaly. These conditions represent a nearly circular low Earth orbit with simplified starting geometry.

The force model uses Earth as the central body, no drag, no solar radiation pressure, no relativistic correction, and an Earth gravity field with degree 4 and order 4 using the JGM2 coefficient file. Propagation is performed with a Runge-Kutta 8/9 integrator, initial step size 60 s, minimum step 0.001 s, maximum step 2700 s, and high accuracy tolerance. The two impulsive burns are defined in the local VNB coordinate system, where the first burn component aligns with the velocity direction. This is appropriate for a Hohmann transfer because the required burns are tangential for the ideal coplanar circular case.

3.1 Selected Simulation Parameters

Parameter	Value Used	Purpose in the Model
Initial semi-major axis	7000 km	Represents the lower near-circular Earth orbit.
Target semi-major axis	12000 km	Creates a clear orbit-raising transfer and visible target altitude.
Initial eccentricity	0.0001	Approximates a circular orbit while avoiding exact-zero numerical sensitivity.
Inclination	28.5 degrees	Provides realistic three-dimensional Earth-orbit geometry.
RAAN, AOP, TA	0 degrees	Simplifies initial orientation so the analysis focuses on altitude change.
Burn 1	0.93497844 km/s	Raises the apoapsis to the target-orbit radius.
Burn 2	0.81612489 km/s	Circularizes the spacecraft at apoapsis.
Burn frame	Local VNB	Applies delta-v in the spacecraft velocity direction.
Final coast	21600 s after Burn 2	Allows the final near-circular orbit to be observed over multiple revolutions.

3.2 Mission Sequence

1. Report the initial orbit state.
2. Apply Burn 1 in the local VNB velocity direction.
3. Propagate until the spacecraft reaches apoapsis.
4. Report the transfer-orbit state before circularization.
5. Apply Burn 2 at apoapsis.
6. Propagate for an additional 21600 s and report the final orbit.

4. Results

The GMAT report file records elapsed time, semi-major axis, and eccentricity throughout the mission. The major state changes match the expected Hohmann-transfer phases: initial near-circular orbit, transfer ellipse after Burn 1, apoapsis arrival, target-orbit circularization after Burn 2, and final near-circular coast.

4.1 Key Reported States

Mission State	Elapsed Time (s)	SMA (km)	Eccentricity	Interpretation
Initial state	0.00	7000.000000	0.00010000	Near-circular lower orbit.
After Burn 1	0.00	9500.319506	0.26325636	Transfer ellipse begins.
At apoapsis before Burn 2	4599.481245	9491.250591	0.26178179	Second burn location.
After Burn 2	4599.481245	11991.693602	0.00131812	Near-circular target orbit.
Final reported state	26199.481244	11989.995111	0.00083411	Final orbit remains near target altitude.

4.2 Analytical Validation

Quantity	Analytical Value	GMAT Value Used/Observed	Difference
Transfer SMA after Burn 1	9500.000000 km	9500.319506 km	+0.319506 km
Transfer eccentricity after Burn 1	0.26315789	0.26325636	+0.00009846
First burn magnitude	0.93497844 km/s	0.93497844 km/s	-3.84e-09 km/s
Second burn magnitude	0.81612489 km/s	0.81612489 km/s	+1.45e-09 km/s
Total delta-v	1.75110333 km/s	1.75110333 km/s	-2.38e-09 km/s
Apoapsis coast time	4607.51 s	4599.48 s	-8.03 s
Final target SMA	12000.000000 km	11989.995111 km	-10.004889 km

The first-burn delta-v in the GMAT script agrees with the analytical value to within approximately 4e-09 km/s, and the second-burn value agrees to within approximately 1e-09 km/s. This confirms that the maneuver magnitudes were not arbitrary values; they were selected from the closed-form Hohmann equations. The final semi-major axis differs from the 12000 km design target by about -10.005 km, corresponding to a relative error of approximately -0.083 percent. The final eccentricity is 0.0008341, which is small enough to classify the achieved orbit as near-circular for this project scale.

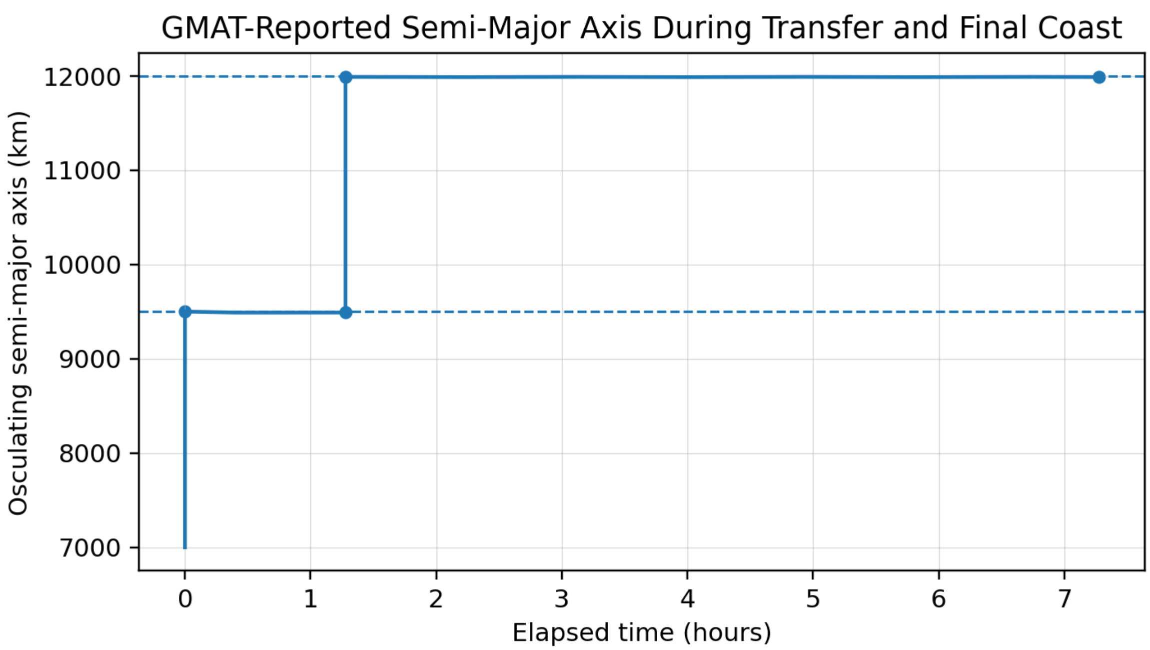

Figure 2. Semi-major-axis history reported by GMAT. The jump after Burn 1 represents injection into the transfer ellipse, and the second jump represents circularization near the 12000 km target orbit.

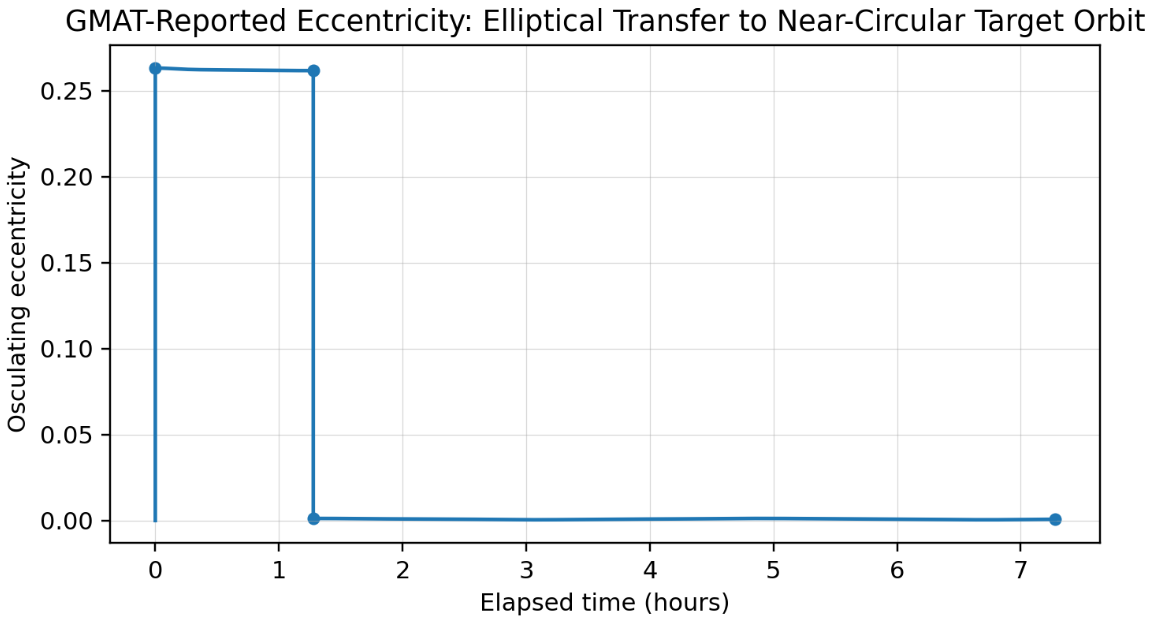

Figure 3. Eccentricity history reported by GMAT. The orbit becomes highly elliptical during transfer and returns to a near-circular state after Burn 2.

5. Discussion

5.1 Physical Interpretation of the Transfer

The simulation demonstrates the central logic of orbital energy control. The initial circular orbit has lower specific orbital energy than the desired final orbit. Burn 1 increases the spacecraft velocity at the lower orbit, increasing the orbital energy and placing the spacecraft onto an ellipse whose apoapsis reaches the target radius. During the coast phase, no additional propulsive work is applied; the spacecraft exchanges kinetic and potential energy as it moves outward. At apoapsis, Burn 2 increases the velocity again so that the local velocity matches the circular-orbit requirement at the target radius.

The reported transition from eccentricity approximately 0.263 after Burn 1 to approximately 0.001 after Burn 2 is the most important numerical signature of a successful Hohmann transfer. It shows that the spacecraft did not simply reach a higher altitude; it also acquired the velocity needed to remain in a stable higher orbit instead of falling back along the transfer ellipse.

5.2 Sources of Small Differences Between Theory and GMAT

The analytical equations assume a two-body point-mass Earth and exactly circular initial and final orbits. The GMAT implementation uses a nonzero initial eccentricity of 0.0001 and propagates osculating elements with a degree/order 4 Earth gravity field. As a result, the semi-major axis and eccentricity are not perfectly constant during the coast phases. This is visible in the report file, where the transfer semi-major axis slowly varies during propagation to apoapsis and the final semi-major axis oscillates slightly around the target value during the post-burn coast.

These deviations do not invalidate the transfer. Instead, they strengthen the project by showing that a realistic numerical tool reports orbital elements as a function of the selected force model and coordinate representation. The achieved final orbit is not mathematically perfect, but it is mission-consistent: it is close to 12000 km and has eccentricity below 0.001 after several hours of propagation.

5.3 Engineering Usefulness of the Model

The project is useful because it creates a complete workflow for early-stage mission design. First, analytical equations provide expected burn values and transfer time. Second, GMAT implements the spacecraft, force model, maneuvers, propagator, and event condition. Third, the output file allows numerical verification. This workflow can be expanded to more advanced investigations such as finite-burn modeling, propellant mass depletion, plane-change optimization, perturbation sensitivity, launch-injection errors, and target-orbit correction maneuvers.

6. Limitations and Future Work

- The model treats the two velocity changes as impulsive burns. Real thrusters operate over finite durations, so a higher-fidelity study should replace the impulses with finite burns and include spacecraft mass depletion.
- The project disables atmospheric drag and solar radiation pressure. These assumptions are acceptable for a first transfer demonstration but should be revisited for low-altitude mission design.
- Only semi-major axis and eccentricity were reported. A stronger validation package would also report periapsis radius, apoapsis radius, velocity magnitude, specific orbital energy, inclination, and true anomaly.
- The target orbit is nearly achieved but not exactly circular. A future version could use GMAT targeting or differential correction to solve for burn values that satisfy final SMA and eccentricity constraints more precisely under the chosen force model.
- The current simulation focuses on a coplanar altitude change. Future work could add a coupled plane change and compare whether splitting the plane change between the two burns reduces total delta-v.

7. Conclusion

This project successfully models and validates a Hohmann transfer from a 7000 km Earth orbit to a 12000 km Earth orbit using GMAT. The analytical Hohmann solution predicts burn magnitudes of 0.93497844 km/s and 0.81612489 km/s, for a total delta-v of 1.75110333 km/s. The GMAT results confirm the expected orbital sequence: a nearly circular initial orbit, an elliptical transfer orbit after the first burn, apoapsis arrival near the theoretical transfer time, and a near-circular final orbit after the second burn. The final reported semi-major axis of 11989.9951 km and eccentricity of 0.0008341 demonstrate that the spacecraft reached a stable orbit very close to the intended 12000 km target. Overall, the simulation provides a technically sound bridge between textbook Hohmann-transfer theory and practical mission-analysis implementation in GMAT.

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